

# Theory Lectures on

## Manufacturing Processes (MEI102)

### ECM Numerical

## EDM – Electric Discharge Machining

(Lecture 7)



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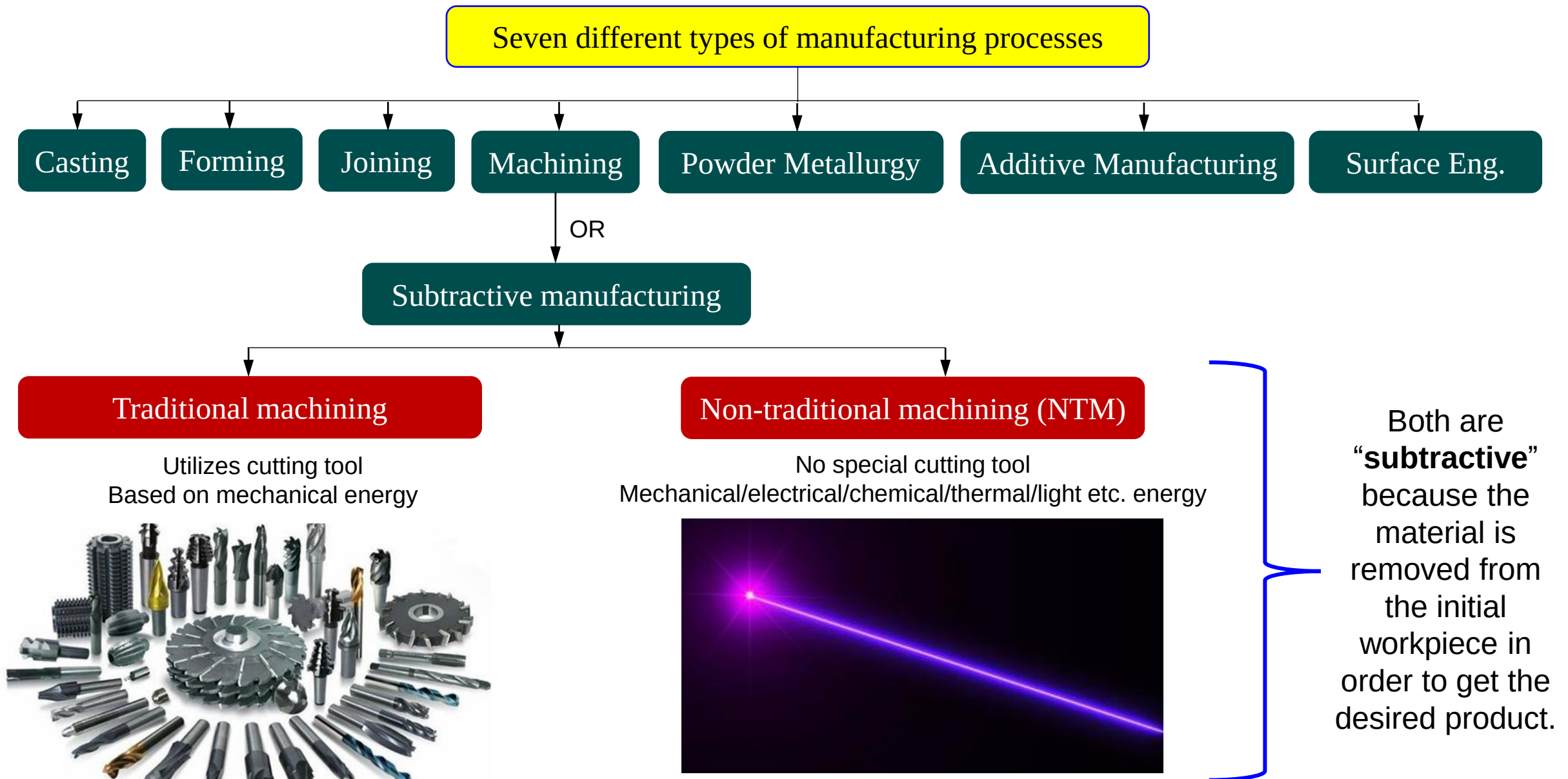
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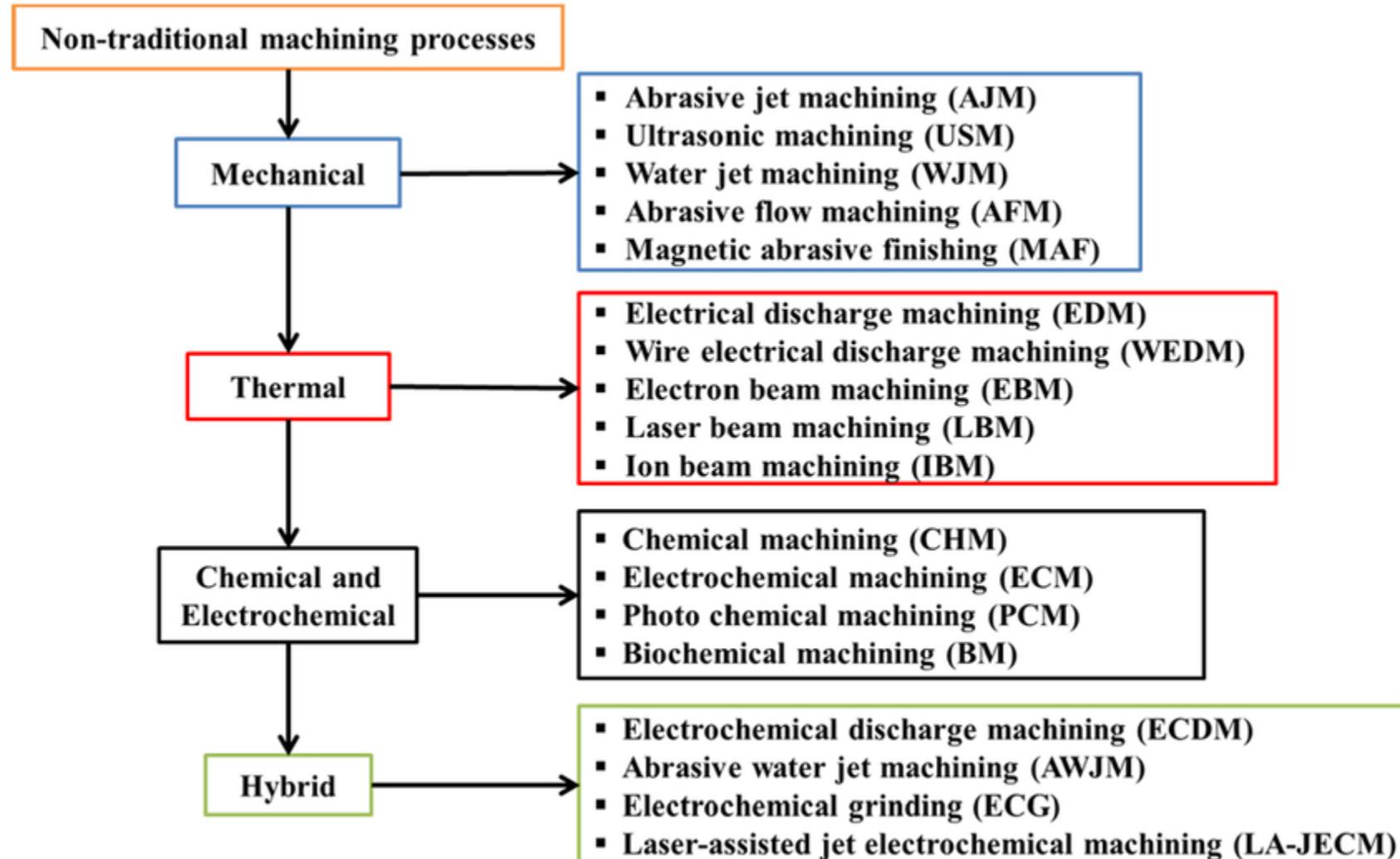
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# Traditional and Non-Traditional Machining

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# ECM Numerical



**Problem:** An alloy consists of 72% nickel, 20% chromium, 5% iron, 0.5% titanium, 0.5% copper, 1.0% silicon, and 1.0% manganese. These are weight percentages. The valency of dissolution is given in following table. Calculate the MRR when this alloy is electro-chemically machined at 500 A current.

| Metal     | Atomic weight (g) | Valency |
|-----------|-------------------|---------|
| Nickel    | 58.71             | 2       |
| Chromium  | 51.99             | 2       |
| Iron      | 55.85             | 2       |
| Titanium  | 47.9              | 3       |
| Copper    | 63.57             | 1       |
| Silicon   | 28.09             | 4       |
| Manganese | 54.94             | 2       |

$$\begin{aligned}
 E_{\text{alloy}} &= \left( \frac{58.72}{2} \times \frac{72}{100} \right) + \left( \frac{51.99}{2} \times \frac{20}{100} \right) + \left( \frac{55.85}{2} \times \frac{5}{100} \right) + \left( \frac{47.9}{3} \times \frac{0.5}{100} \right) \\
 &+ \left( \frac{63.57}{1} \times \frac{0.5}{100} \right) + \left( \frac{28.09}{4} \times \frac{1}{100} \right) + \left( \frac{54.94}{2} \times \frac{1}{100} \right) = 28.473 \text{ gm}
 \end{aligned}$$

$$\therefore MRR = \frac{1}{F} IE = \frac{1}{96500} \times 500 \times 28.473 = 0.1475 \text{ gm/s}$$

**Governing Equation-1:** Faraday's Law of electro-chemical dissolution states that the mass of material removed is proportional to the charge and the same is also proportional to the electrochemical equivalent.

$$m \propto Q \text{ and } m \propto E$$

$$\therefore m = \frac{1}{F} ItE$$

$$\therefore MRR = \frac{m}{t} = \frac{1}{F} IE$$

$F$  = Faraday's Constant = 96500 A.s

$I$  = Current in A.

$t$  = Time of dissolution in s.

**Governing Equation-2:** Electro-chemical equivalent ( $E$ ) of an alloy is calculated by multiplying the chemical equivalent of individual elements by their respective portions by weight, and then summing up, as given in the following expression

$$E_{\text{alloy}} = \left( \frac{M}{V} \right)_{\text{alloy}} = \sum_{i=1}^n \left( \frac{M_i}{V_i} \right) \frac{X_i}{100}$$

$M$  = Atomic weight in gram.

$V$  = Valency.

$X$  = Weight portion in %.



**Problem 1:** Titanium alloy (Ti-6Al-4V) is electro-chemically machined at a current of 2000 A. The composition (% by weight) of this alloy is given in the following table. Calculate the MRR.

| Element           | Ti   | Al    | V     | Fe    | Cu    |
|-------------------|------|-------|-------|-------|-------|
| Weight % in alloy | 89   | 6.1   | 4     | 0.5   | 0.4   |
| Atomic Weight (g) | 47.9 | 26.97 | 50.94 | 55.85 | 63.57 |
| Valency           | 3    | 3     | 4     | 2     | 1     |

**Problem 2:** Turbine blade made of nickel alloy is electro-chemically machined at a current of 1500 A. The composition (% by weight) of this alloy is given in the following table. Calculate the MRR.

| Element           | Co   | Cr   | Ni      |
|-------------------|------|------|---------|
| Weight % in alloy | 18   | 20   | Balance |
| Atomic Weight (g) | 59.9 | 52.0 | 58.7    |
| Valency           | 2    | 6    | 2       |

Hint: Total weight % of individual elements of any alloy has to be 100. Therefore, **balance** indicates 100 minus rest of the elements. In this case, Ni weight is  
 $100 - 18 - 20 = 62 \%$ .

# **Electric Discharge Machining (EDM)**



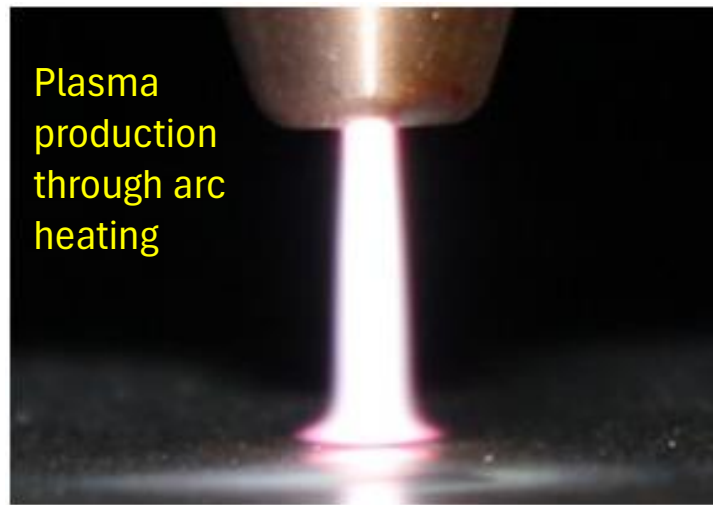


# What is an electric arc?

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Arc Welding

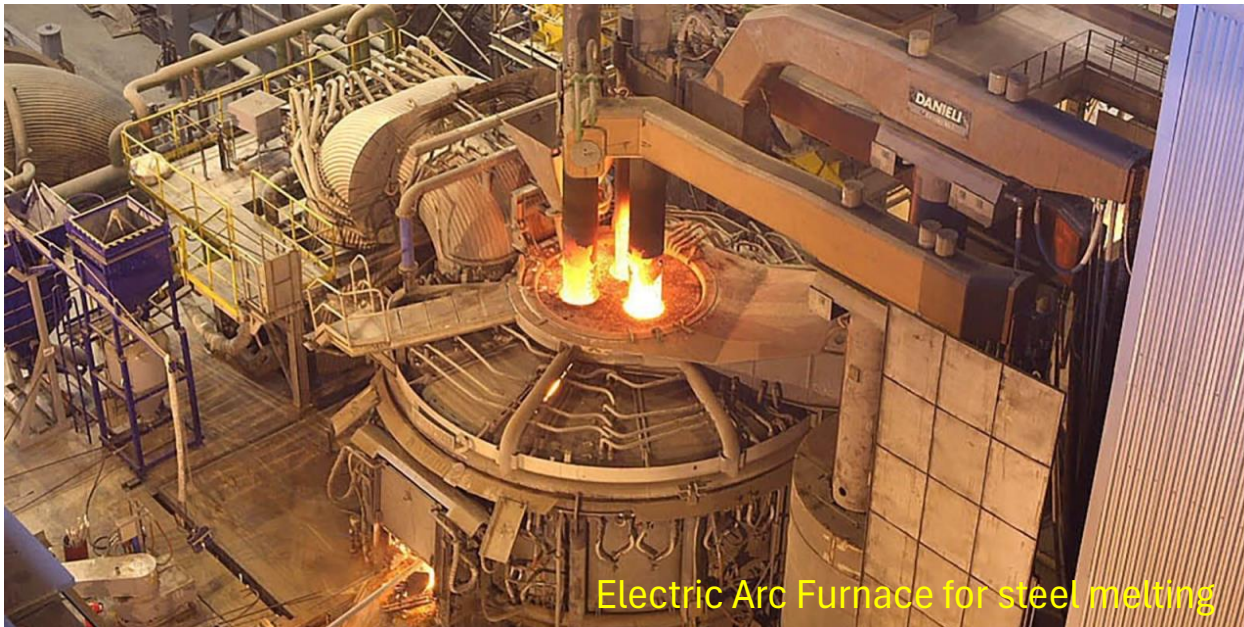


Plasma production through arc heating

**Electric arc:** Continuous flow of avalanche of electrons in between two electrodes through a narrow conductive channel surrounded by a non-conductive medium. Here the narrow conductive channel sustains for longer duration.

## Arc formation mechanism

1. Initially two electrodes are separated by a non-conductive medium (such as air or dielectric fluid).
2. When sufficient voltage is applied across two electrodes, a few electrons attempt to flow from one electrode to another overcoming the barrier of non-conductive medium.
3. This causes collision between the electrons and atoms of the non-conductive medium. As a result, the atoms of the non-conductive medium get ionized.
4. Eventually, a narrow column of this non-conductive medium ionizes resulting in the formation of a conductive channel.
5. A larger number of electrons then flow through this channel resulting in the formation of an arc. The voltage and other parameters must be sufficient to sustain this electron flow for a longer duration.



Electric Arc Furnace for steel melting

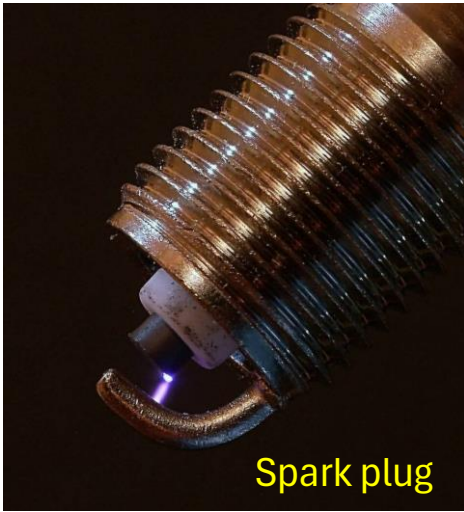




# What is an electric spark?

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Electric Spark



Spark plug



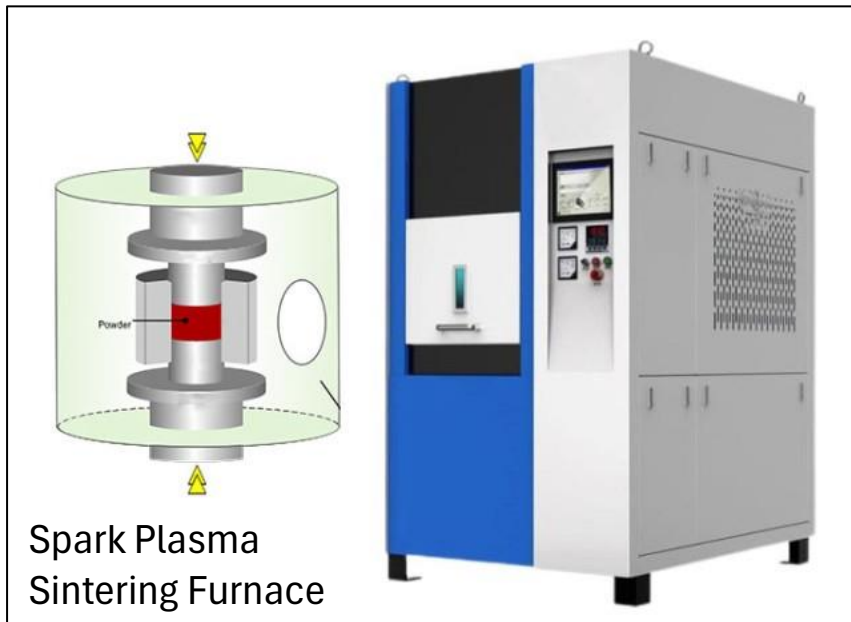
Natural lightning

**Electric spark:** Intermittent flow of avalanche of electrons in between two electrodes through a narrow conductive channel surrounded by a non-conductive medium. Here the narrow conductive channel forms only for a short period of time.

Formation mechanism of arc and spark are fundamentally same; however, spark forms only for a short duration of time. Voltage, inter-electrode gap, or other process parameters are controlled in such a way that prohibit prolonged formation of conductive channel through the non-conductive medium. In other words, conductive channel does not sustain for a longer duration during spark formation (on contrary, conductive channel sustains for a longer duration in arc).

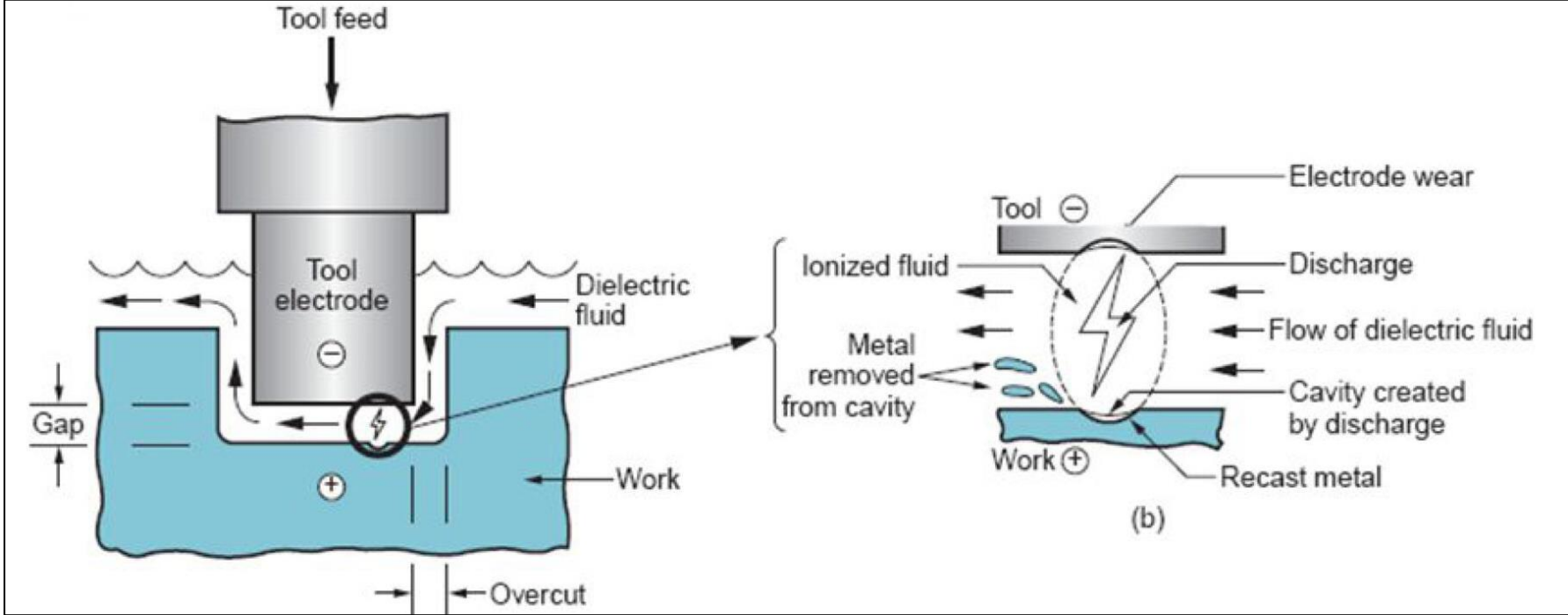
This spark is used in EDM for material removal.

In EDM, the non-conductive medium is a suitable dielectric fluid (such as kerosene, de-ionized water, etc.).



Spark Plasma  
Sintering Furnace

## Working principle of EDM

- EDM is one thermal energy based process where intense heat of the spark is used to locally melt and vaporize the workpiece.
  - A shaped-tool is brought close to the workpiece maintaining a small inter-electrode gap (IEG). Both the tool and workpiece are connected to electrical circuit (both tool and workpiece must be conductive).
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- The diagram illustrates the EDM process. On the left, a cross-sectional view shows a 'Tool electrode' (labeled with a minus sign) moving downwards, indicated by a 'Tool feed' arrow. It is positioned above a 'Work' piece (labeled with a plus sign). The space between them is the 'Gap', and the fluid filling this space is 'Dielectric fluid'. An 'Overcut' is shown at the bottom of the gap. On the right, a detailed view of the discharge process is shown, labeled '(b)'. It depicts a 'Discharge' (represented by a lightning bolt) between the 'Tool' (labeled with a minus sign) and the 'Work' (labeled with a plus sign). This process results in 'Electrode wear' on the tool, 'Metal removed from cavity' from the workpiece, and 'Recast metal' on the workpiece surface. The 'Flow of dielectric fluid' is shown entering the gap, and 'Ionized fluid' is shown exiting. A 'Cavity created by discharge' is also indicated.
- A suitable dielectric fluid is continuously flushed through the IEG (i.e. between tool-electrode and workpiece-electrode). This dielectric fluid is electrically non-conductive fluid in normal condition.
  - When sufficient voltage is applied across two electrode, localized **dielectric breakdown** occurs resulting in the formation of a narrow conductive channel of ions for a short duration (few micro-seconds). Avalanche of electrons thus flow through this conductive channel resulting in the formation of a spark. As the accelerated electrons strike the work surface, their kinetic energy is transformed to thermal energy resulting in very high localized temperature generation (5,000 – 10,000 °C).
  - Spark heat instantly melts and vaporizes a tiny volume of material from the workpiece surface forming a small crater. The removed material (debris) is flushed away by the flowing dielectric medium. Repeated formation of spark results in significant volumetric removal.



## The electrode should possess several important properties:

- High electrical conductivity – electrons are cold emitted more quickly, and there is less bulk electrical heating.
- High thermal conductivity – for the same heat load, the local temperature rise would be less due to faster heat conducted to the bulk of the tool and thus minor tool wear..
- Higher density – for the same heat load and same tool wear by weight, there would be less volume removal or tool wear and thus less dimensional loss or inaccuracy.
- High melting point – high melting point leads to minor tool wear due to less tool material melting for the same heat load.
- Easy manufacturability – tool is shaped-tool and thus specific complex shape is to be given on the tool.

The followings are the different electrode materials that are used commonly in the industry

- Graphite
- Electrolytic oxygen-free copper
- Tellurium copper – 99% Cu + 0.5% tellurium
- Brass

Note – In EDM, electrode material is independent of the workpiece material. This is unlike ECM which require judiciously selected electrode material based on the workpiece material.



## The dielectric fluid should possess several important properties:

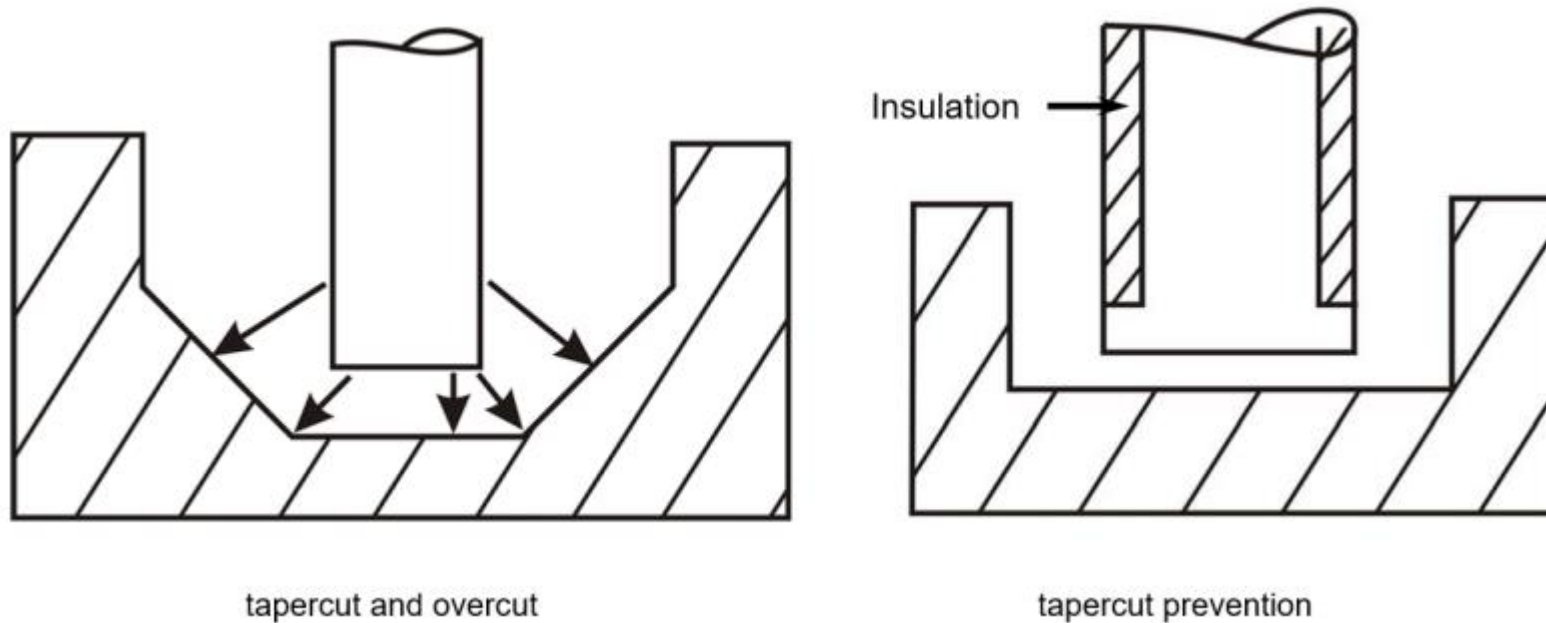
- Dielectric fluid should provide an oxygen free machining environment (otherwise high-temperature oxidation of the work surface can occur).
- It should have enough strong dielectric resistance so that it does not breakdown electrically too easily but at the same time ionize when electrons collide with its molecule.
- During sparking it should be thermally resistant as well.
- It should be non-toxic.
- It should also be chemical non-reactive with the tool and workpiece material.

Generally **kerosene** and **deionized water** are used as dielectric fluid in EDM. **Tetraethylene glycol** is also often used for precision EDM machining.

Tap water cannot be used as it ionizes too early and thus breakdown due to presence of salts as impurities occur.

Dielectric medium is generally flushed around the spark zone.

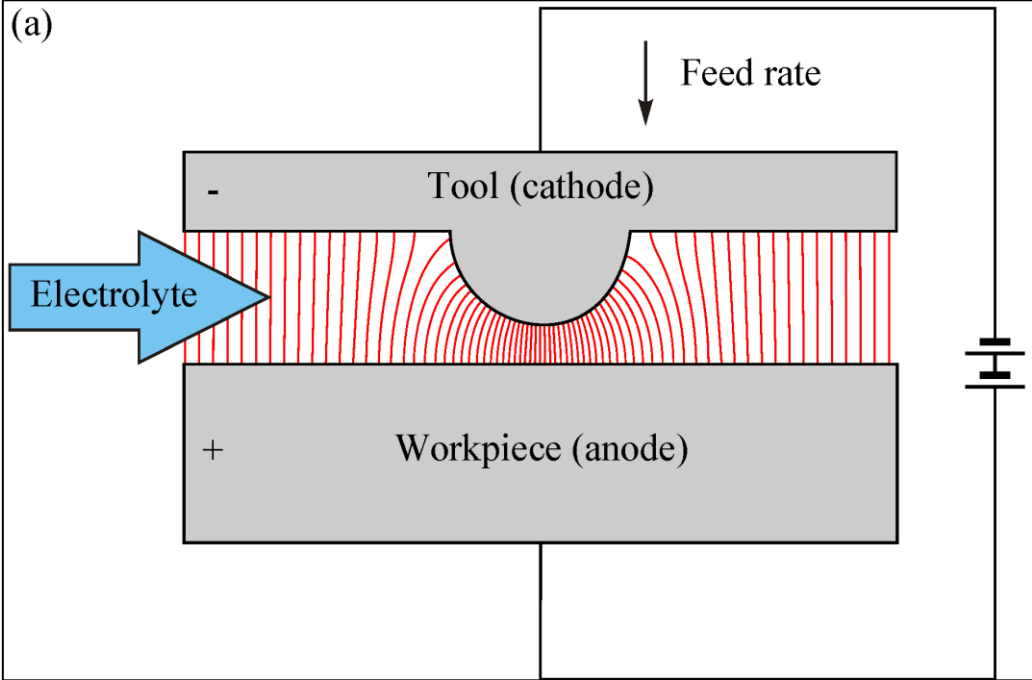
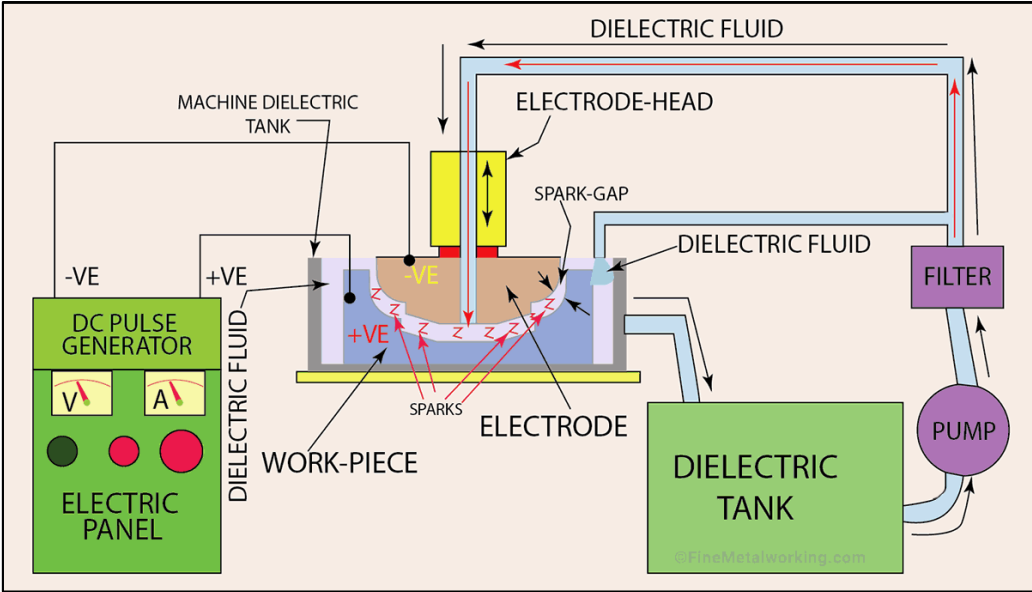
In EDM, dielectric fluid is independent of the workpiece material. This is unlike ECM which requires appropriate electrolyte based on the workpiece material for electro-chemical reaction to take place.



- In EDM, the spark occurs between the two nearest points on the tool and workpiece (lowest length between two electrodes, so lowest resistance, so easiest to breakdown and conductive channel formation).
- Thus, machining may occur on the side surface as well, leading to overcut and taper cut.
- Taper cuts can be prevented by suitable insulation of the tool. Overcut cannot be prevented as it is inherent to the EDM process. But the tool design can be done in such a way, so that same gets compensated.



# Difference between Dielectric (EDM) and Electrolyte (ECM)



| Dielectric                                                                                                                                             | Electrolyte                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dielectric fluid is utilized in electric-discharge machining (EDM) process.                                                                            | Electrolyte is utilized in electro-chemical machining (ECM) process.                                                                                                |
| A dielectric should be electrically non-conductor at normal condition; however, it should behave as good conductor after <i>dielectric breakdown</i> . | Electrolyte should always be good conductor of electricity for completing the electrical circuit to allow smooth flow of electrons and ions between two electrodes. |
| The dielectric should be oxygen free. Due to extreme spark heat, liberation of oxygen can oxidize the work surface.                                    | Presence of oxygen in electrolyte does not matter as ECM process does not generate extreme heat.                                                                    |
| Common dielectric fluids used in EDM include deionized water, transformer oils, kerosene, paraffin, etc.                                               | Commonly used electrolytes in ECM process include sodium chloride (NaCl) and sodium nitrate (NaNO3) solutions.                                                      |





# Difference between EDM and ECM processes

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| Electro-Discharge Machining (EDM)                                                                                                                                                                | Electro-Chemical Machining (ECM)                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EDM is one <i>thermal energy</i> based non-traditional machining process.                                                                                                                        | ECM is <i>electro-chemical energy</i> based non-traditional machining process.                                                                                       |
| Material removal takes place by <i>melting and vaporization (spark erosion)</i> . Thus material gets removed in molten and vaporized state.                                                      | Material is removed by controlled <i>electro-chemical dissolution</i> of work metal (anode). Material removal takes place <i>in the form of ions of work metal</i> . |
| Owing to electric spark, significant <i>heat is generated</i> in EDM process. The localized temperature reaches as high as <i>10,000°C</i> .                                                     | <i>No heat generation</i> takes place in ECM. Accordingly, temperature at the machining zone does not increase.                                                      |
| An appropriate <i>dielectric fluid</i> (such as kerosene, paraffin, deionized water) is used in EDM to assist spark generation.                                                                  | A suitable <i>electrolyte</i> (such as NaCl and NaNO <sub>3</sub> ) is applied in ECM to assist flow of ions between two electrodes.                                 |
| During EDM operation, <i>electrode material is also removed gradually</i> along with workpiece material. However, electrode material removal rate is much lower than work material removal rate. | <i>Electrode does not suffer any wear</i> during ECM process. Same electrode can be used repeatedly for a longer duration. Accordingly, electrode life is higher.    |
| EDM process typically requires <i>high potential difference (50 – 400V) and moderate current (100 – 200A)</i> .                                                                                  | ECM process requires <i>moderate voltage (&lt;50V) but very high current (&gt;1000A)</i> .                                                                           |



## Advantages of EDM

- EDM can efficiently cut very hard materials (superalloys) as long as it is electrically conductive.
- Tight tolerances and intricate shapes with exceptional precision can be achieved.
- Machinability of the work material is independent of its mechanical, optical, and chemical properties.
- Machined surfaces are free from burrs
- No harmful chemical is used.

## Disadvantages of EDM

- Applicable to electrically conductive material only.
- Tool-electrode undergoes gradual wear; thus its shape changes with time.
- Vertical walls cannot be produced due to sparking on the side walls.
- Workpiece material experiences thermal damages at the machined surface.

## Applications of EDM

- Cutting tool fabrication.
- Cutting and texturing very hard materials and superalloys.
- Drilling holes through the hard nozzles.
- Die and mould making
- Precision finishing of very hard materials.
- Micro-texturing for hydrophobic surface generation.